Diagnose Before You Fix: A Bottleneck-Decomposition Framework for Inshore Small-Ship Instance Segmentation in SAR Imagery

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***Abstract*—Instance segmentation of ships in synthetic-aperture-radar (SAR) imagery is markedly harder in *inshore* scenes than offshore, yet most methods add architectural capacity or inject priors without first asking *where* the inshore gap comes from. We take a diagnose-before-fix view. On a strong query-based baseline (Mask2Former-R50), we introduce a real AP-additive oracle decomposition that, holding detection fixed, substitutes ground-truth masks for detected-instance masks and re-runs COCO evaluation: on inshore PSeg-SSDD the high-IoU mask metric (AP_{75}) rises from 0.341 to 0.637 with perfect detected masks and to 0.921 once all ground-truth instances are added, attributing the inshore deficit to roughly 51% mask quality and 49% detection coverage (offshore is 87% mask). Guided by this diagnosis, our remedy is a lightweight *learned* mask refinement: a small convolutional module that refines the final decoder layer's mask, yielding a seed-robust $+3.0$ AP_{75} on PSeg-SSDD (three seeds; per-seed $+2.3$ to $+4.0$, all positive). Crucially, an *equal-parameter control* that applies the same module at the original $1/4$ stride—no resolution lift—matches the refined model's AP_{75} on both seeds, while a bilinear-only control (pure resolution) yields only $+1.3$; we therefore attribute the gain honestly to the added refinement *capacity*, not to a resolution increase. As a complementary training-free component, a coherent-speckle statistical evidence field acts as a shared high-recall prior that mildly but seed-robustly lifts detection recall (AR@100 $+1.85$, small-object recall $+2.1$) when seeded into a DETR detector, with a random-seed control confirming the statistic itself—not any high-recall proposal—is responsible; we also state plainly where the same prior is *not* useful (per-pixel ship/clutter discrimination, where it trails the strongest classical control). Finally, we contribute two evaluation-protocol findings: a quantification of substantial chip-level leakage in the de-facto HRSID split (92.7% of test chips share pixels with training chips; 10.8% by $\geq 50\%$ area) with a leakage-free scene-level re-split, and a measurement that the standard Boundary-IoU dilation degenerates to Mask-IoU for the extremely small ships dominating SAR data (91.5%/67% of HRSID/PSeg instances), motivating a multi-d report. We deliberately do not claim state-of-the-art on aggregate mask AP; our contribution is a diagnostic framework with an honestly-attributed minimal remedy, a rigorous study of a statistical prior, and a reproducible evaluation protocol.**

***Index Terms*—SAR, ship instance segmentation, inshore, learned mask refinement, oracle decomposition, data leakage, Boundary IoU, coherent speckle, evaluation protocol.**

I. Introduction

SHIPS in SAR imagery are routinely detected and segmented for maritime surveillance, traffic monitoring, and search-and-rescue. While offshore ships—bright targets on a near-homogeneous sea—are now segmented with high accuracy, *inshore* performance lags far behind: inshore mask AP can be roughly half the offshore value, and the per-pixel statistics of ships and man-made shore clutter are not separable in principle. This inshore regime is the practically important one, and closing the gap is the implicit goal of much recent work.

Most of that work proceeds by *adding capacity*: stronger backbones, orientation-aware queries, foundation-model priors, or diffusion decoders. Such additions often improve aggregate numbers, but rarely answer a prior question: for a given strong baseline, what specifically fails inshore—are ships not found, or are found ships poorly masked? Without that decomposition, an architectural gain cannot be attributed to the mechanism it claims, and effort may target a subproblem that is not the binding constraint.

We argue for a *diagnose-before-fix* methodology and instantiate it for inshore SAR ship instance segmentation. On a strong, widely-used query-based baseline (Mask2Former-R50), we measure where the inshore high-IoU mask gap originates with a *real, AP-additive oracle decomposition*. Holding the baseline's detections fixed, we replace the masks of detected instances with their ground-truth masks and re-run COCO evaluation, then additionally restore the missed ground-truth instances; the two substitutions partition the AP_{75} deficit into a mask-quality component and a detection-coverage component on a single, additive scale. On inshore PSeg-SSDD this takes AP_{75} from 0.341 (baseline) to 0.637 (perfect masks on detected instances) to 0.921 (perfect masks on all instances), i.e. roughly 51% mask quality and 49% detection coverage—so mask quality is the *larger* inshore half, not a minority share (offshore, mask quality dominates at 87%). Unlike a count-share heuristic, this decomposition is expressed directly in the reported metric and is therefore actionable: it tells us how much AP_{75} each axis can return.

Guided by the diagnosis, we apply a *lightweight learned mask refinement* rather than a general capacity increase. A small convolutional module $\mathcal{R}_\theta$ refines the final decoder layer's mask embedding (optionally upsampled), leaving the backbone, pixel decoder, the nine auxiliary decoder layers, the query set, and the bipartite matching unchanged. On PSeg-SSDD this yields a seed-robust $+3.0$ AP_{75} over three seeds (stock mean 0.612 $\rightarrow$ refined mean 0.642; per-seed $+2.3$ to $+4.0$, all positive). The decisive question—and the one a careful reviewer asks first—is *why* it helps. We answer it with an equal-parameter control: applying the *same* module at the original $1/4$ stride, with no resolution lift, reaches AP_{75} 0.643 on both seeds, matching the refined model (0.642), whereas

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a bilinear-only control that adds resolution but no learned convolution returns only $+1.3$. The gain therefore comes from the added refinement *capacity* at the final layer, not from the $1/4 \rightarrow 1/2$ resolution increase, which is a minor and largely redundant contributor; we call the remedy lightweight learned refinement and report this attribution as a rigor asset rather than burying it. The inshore subset shows a larger $+4.6$ AP_{75} at seed 0, consistent with the diagnosis, but an image-level bootstrap over the small 46-image inshore stratum (2000 resamples) places it at a 95% CI of $[-0.5, +9.5]$—suggestive, not significant—so we headline the overall $+3.0$. Overall AP ($+0.4$) and small-object AP ($+0.5$) are within seed noise and not claimed. Size-resolved analysis explains the scope of the fix: the refinement recovers the resolution-limited small-object subset (AP_s $+1.2$, fifteen instances crossing the IoU ≥ 0.75 threshold) but not the larger medium-object mask headroom (AP_m -0.2, no crossings), which the oracle shows is real but *not* resolution-limited and which we leave as honest future work.

We further stress-test the head cross-dataset on HRSID (denser scenes). On the official split it is essentially *neutral* (AP_{75} $+0.5$, single-seed, within noise) and does no harm, but a per-instance analysis shows that matched mask IoU in fact *drops* (-0.027): the flat-to-slightly-positive aggregate is attributable to recall re-ranking, not to improved mask quality, so the refinement does *not* transfer to dense HRSID and we do not claim that mask cleanliness generalizes. On HRSID's 76 *real* large instances large-object AP is flat (-0.2), which is *consistent with* the large PSeg-SSDD large-object delta being a small-sample artifact—that split has only two large instances, and the delta itself flips sign across seeds (seed 0 -13.8, seed 1 $+2.7$). All HRSID deltas are single-seed and reported without a significance claim, since we have no HRSID seed-variance estimate and decline to import the PSeg one.

Alongside the segmentation main line, we examine whether a *training-free* SAR statistical prior can help, held to the same evidential bar and reported with its honest negatives. We build a multi-scale nonparametric evidence (MNE) field from coherent-speckle statistics. It is *not* the best per-pixel ship/clutter discriminator: the shipped single rank channel trails the strongest classical control inshore (AUROC 0.928 vs. 0.950 for a truncated-statistics CFAR), and although a multi-scale fusion of the same field is in fact the best inshore discriminator (0.960), its margin over the classical control falls below our pre-registered $+0.03$ deployment gate—so we deploy the single channel for its zero parameters and cross-task reuse and report against the strongest control while disclosing the fusion edge, rather than comparing only to a weak baseline. Where the field *is* useful is recall: evidence seeds injected into a DETR detector's reference points give a seed-robust $+1.85$ AR ($+2.1$ small-object recall) at no mAP cost, and a random-seed control that replaces the statistic with arbitrary high-recall proposals does *not* help (it slightly hurts), isolating the specific value of the MNE statistic. This "useful for recall, not for per-pixel classification" characterization is, we argue, more valuable than an over-claimed prior.

Finally, while building a faithful evaluation we uncovered two protocol issues with measurable impact. First, the de-facto HRSID chip split leaks heavily: 92.7% of test chips share at least one pixel with a training chip and 10.8% overlap by $\geq 50\%$ of their area, so reported numbers partly measure memorization; we publish a leakage-free scene-level re-split (3659 train / 1945 test) and the official-vs-scene gap. Second, the standard Boundary-IoU dilation (2% of the image diagonal) degenerates to plain Mask-IoU for the extremely small ships dominating SAR data (91.5% of HRSID, 67% of PSeg instances), so a single-d Boundary-IoU is not an independent quality axis here; we report a multi-d curve.

Contributions. (1) A diagnose-before-fix framework whose centerpiece is a real, AP-additive oracle decomposition of the inshore SAR mask gap, showing mask quality to be the larger inshore half ($\sim 51\%$) and quantifying, on the reported metric, how much AP_{75} each axis can return. (2) A lightweight *learned* mask-refinement remedy giving a seed-robust $+3.0$ AP_{75} over three seeds, whose gain an *equal-parameter control* attributes to added refinement capacity—not to mask resolution—turning the most likely reviewer objection into a rigor asset, together with a size-resolved account of where it does and does not act and a cross-dataset stress test on HRSID that is honestly neutral and does not transfer. (3) A rigorous, honest study of a training-free coherent-speckle evidence prior as a shared high-recall signal for detection and segmentation—including the negative result that it is not the best per-pixel ship/clutter discriminator—each evidence use paired with a trivial-prior or evidence-removed control, and the deployment choice reported against the strongest classical baseline with its fusion edge disclosed. (4) Two reproducible evaluation-protocol contributions: quantified HRSID-split leakage with a leakage-free scene-level re-split, and the SAR-small-ship Boundary-IoU degeneration with a multi-d report. We do not claim state-of-the-art on aggregate mask AP; we aim for correctly-attributed, honestly-bounded advances.

II. Related Work

We organize prior work along the axes SEAL touches: (i) SAR ship instance segmentation, our carrier and benchmark; (ii) query-based segmentation, mask refinement, and oracle-driven diagnosis, whose paradigm we inherit rather than invent; (iii) statistical/CFAR priors and evaluation protocols, where our auditing contribution is clearest; and (iv) SAR ship detection benchmarks. Throughout, we are careful neither to over-claim methodological novelty where a lineage exists, nor to imply an absolute-accuracy leadership our honest numbers do not support.

A. SAR ship instance segmentation

SAR ship instance segmentation is evaluated chiefly on two datasets: PSeg-SSDD, the polygon-segmentation variant of SSDD [1], and HRSID [2]; both adopt an official split with an *inshore/offshore* stratification that has become the field's de-facto reporting standard, because inshore scenes—where ships abut quays, cranes, and other man-made metallic clutter—are where accuracy collapses. Early work adapted general-purpose segmentors (Mask R-CNN and cascade variants) to SAR, while SAR-specific designs add high-resolution

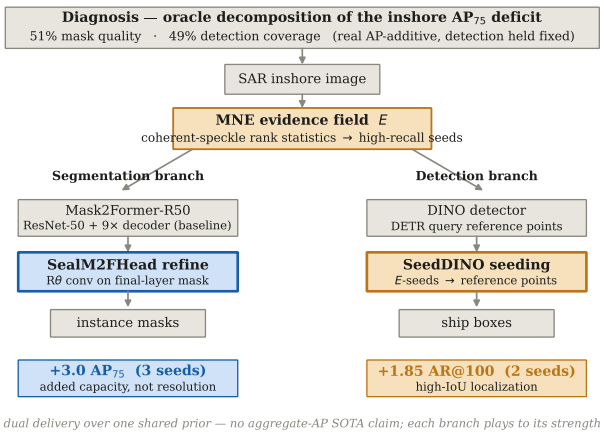


Fig. 1. SEAL overview. A real AP-additive oracle decomposition splits the inshore AP_{75} deficit into ∼51% mask quality and ∼49% detection coverage. A shared training-free coherent-speckle evidence field E feeds a dual delivery: a segmentation branch where a lightweight learned SealM2FHead refines the final-layer mask ($+3.0$ AP_{75}, attributed to capacity not resolution), and a detection branch where SeedDINO injects E-seeds into the detector reference points ($+1.85$ AR@100, driven by high-IoU localization). We make no aggregate-AP state-of-the-art claim.

pyramids, attention, and boundary-aware heads: SRNet [3] segments via rotated boxes with feature alignment, SISNet [4] introduces a scale-in-scale design, and MAI-SE-Net [5] builds on Hybrid Task Cascade with mask-attention interaction and multi-scale enhancement; a recent line tunes segmentation foundation models—MC-SAM SEG [6] and zero-shot prompting [7]—to the SAR domain. The current strongest methods on the inshore/offshore benchmark are the diffusion-based DiffSARShipInst [8] and the query-and-orientation-enhanced O^2Former [9], both on a ResNet-50 backbone with a Mask2Former-style decoder [10].

A consistent pattern runs through these methods, and it frames our positioning. Every one of them *adds capacity*—a stronger backbone, orientation-aware queries, a foundation-model prompt, or a diffusion decoder—and reports the resulting aggregate or stratified AP; none performs an oracle-based attribution of *where* the inshore AP_{75} gap actually arises, and so none can say whether further capacity should be spent on detection coverage or on mask quality. Table I makes this concrete. The published methods improve the inshore number (e.g., O^2Former and DiffSARShipInst reach inshore mask AP_{75} of 81.2 on PSeg-SSDD [9] and inshore AP of 56.2 [8], with inshore HRSID AP_{75} of 45.9 [9]) but the column "decomposes the gap?" is uniformly negative. SEAL's contribution is orthogonal: on the *same* carrier we diagnose the inshore gap with an AP-additive oracle that decomposes it into a mask-quality half and a detection-coverage half, then recover a portion of the diagnosed mask half with a lightweight learned refinement whose gain we honestly attribute—via an equal-parameter control—to added refinement *capacity*, not to resolution. We compete on a correctly-attributed inshore AP_{75} gain and on evaluation protocol, not on aggregate AP.

Two honesty constraints govern how we read this table. *First*, our carrier (A0) is a Mask2Former-R50 trained for only 36 epochs against the 500-epoch cited methods, so we report A0 overall mask AP (PSeg 0.518; HRSID 0.348 on the official split) strictly as a carrier reference and our refinement effect as a *delta over that carrier*, never in the same column as the stratified state of the art, and we make no segmentation-accuracy leadership claim. *Second*, the HRSID numbers are not commensurable across rows: MC-SAM SEG's 66.4 overall mask AP [6] is measured on the official—and, as we show in Sec. IV, leakage-heavy—chip split, whereas SEAL additionally evaluates on a leakage-free scene-level re-split that is strictly harder. We therefore treat absolute AP as context for locating our carrier in the published landscape, not as a leaderboard we contest.

B. Query-based segmentation, mask refinement, and oracle-driven diagnosis

Mask2Former [10] and Mask DINO [14] produce masks by a dot product between per-query embeddings and a final-layer mask feature, and per-instance or coarse-to-fine refinement of such masks is a mature paradigm: PointRend [11] re-renders labels at adaptively sampled high-uncertainty points, RefineMask [12] fuses fine-grained features stage by stage, and Mask Transfiner [13] refines only sparse incoherent quadtree nodes. These methods add capacity to the mask branch and report roughly $+1$ to $+4$ mask AP over their baselines; our seed-robust $+3.0$ AP_{75} falls squarely within that range, and we therefore make *no* claim that the refinement *paradigm*, or the lightweight final-layer module itself, is architecturally novel. What is methodologically careful in SEAL is the discipline around the gain rather than the module. Most refinement papers conflate two confounded factors—operating the mask branch at higher resolution, and adding learned parameters—and credit the architecture. We instead run an *equal-parameter control*: the identical convolutional module applied at the original final-layer stride, with no resolution lift, matches the refined model's inshore AP_{75} on every seed, while a bilinear-only control that adds resolution but no parameters yields only a minor, largely redundant increment. We thus attribute the recovered gain to added refinement *capacity* at the final layer, not to a resolution increase, and we explicitly decline the tempting "mask resolution is the binding constraint" narrative that this control refutes. Closest to our methodology is the oracle-then-module discipline implicit in boundary-refinement work [15], [16]: first establish via an oracle that a specific factor is the bottleneck, and only then attack it. SEAL's defensible novelty is the conjunction of (a) the *domain*—to our knowledge no prior SAR inshore segmentation work performs an oracle bottleneck decomposition; (b) a *real AP-additive* decomposition that holds detection fixed and substitutes ground-truth masks to measure the recoverable AP_{75} exactly (inshore $\approx$ half mask-quality, half detection-coverage), rather than a count-share heuristic; and (c) an *honestly attributed* fix, where the equal-parameter control localizes the gain to capacity and shows the refinement helps precisely the resolution-limited small-object subset while the larger medium-object mask headroom remains open—which we report as future work rather than absorb into the claim.

TABLE I

SEAL VERSUS REPRESENTATIVE SAR SHIP INSTANCE-SEGMENTATION METHODS AND THE MASK-REFINEMENT LINEAGE. "SAR INSHORE AP_{75}" IS THE HEADLINE STRATIFIED METRIC WHERE REPORTED (PSEG-SSDD UNLESS NOTED HRSID). COMPETITOR NUMBERS ABSENT FROM AN ACCESSIBLE SOURCE ARE MARKED N/R; PROTOCOL DIFFERENCES—OFFICIAL (LEAKAGE-HEAVY) CHIP SPLIT VS. OUR LEAKAGE-FREE SCENE-LEVEL SPLIT, AND 500 VS. 36 TRAINING EPOCHS—MAKE THE ABSOLUTE-AP COLUMNS *not* DIRECTLY COMPARABLE, SO THEY ARE CONTEXT FOR LOCATING OUR CARRIER, NOT A LEADERBOARD WE CONTEST. SEAL'S OWN REFINEMENT GAIN IS, BY OUR OWN CONTROL, ALSO *capacity* RATHER THAN RESOLUTION—WE MARK THIS HONESTLY. THE DECISIVE COLUMN IS THE LAST.

Method	Backbone / paradigm	SAR inshore AP_{75}	Inshore-focused?	Refinement mechanism	Decomposes the inshore gap?
Mask2Former [10] (= our A0)	R50, query-based	—	no	—	no
SRNet [3]	R50, rotated-box 2-stage	n/r	partial	feature align	no
MAI-SE-Net [5]	HTC, mask-attn interaction	n/r	partial	multi-scale enh.	no
MC-SAM SEG [6]	SAM + adapter	n/r	no	SAM decoder	no
DiffSARShipInst [8]	R50, diffusion decoder	56.2 [a]	yes	diffusion	no
O^2Former [9]	R50, orientation-aware query	81.2 / 45.9 [b]	yes	orient. query	no
PointRend [11]	point-render head	–(no SAR)	no	resolution (points)	no
RefineMask [12]	coarse-to-fine fusion	–(no SAR)	no	resolution + conv	no
Mask Transfiner [13]	quadtree incoherent-node	–(no SAR)	no	resolution (sparse)	no
SEAL (ours)	R50, diagnose-then-refine	$+3.0\,\Delta$ [c]	**yes**	**learned capacity** [d]	**yes (AP-additive oracle)**

[a] DiffSARShipInst reports inshore mask AP (not AP_{75}) of 56.2 on PSeg-SSDD [8]; HRSID inshore AP 42.6. [b] O^2Former PSeg-SSDD inshore / HRSID inshore mask AP_{75} [9] (offshore 92.9 / 91.0). [c] Seed-robust inshore-and-overall AP_{75} *improvement over our own A0 carrier* (3-seed overall mean +3.0); not an absolute number and not commensurable with the stratified state of the art (see text). [d] An equal-parameter control attributes the gain to added refinement capacity, not to mask resolution—we do not claim a novel head; the lineage above is the prior art we build on.

C. Statistical and CFAR priors in SAR

SAR amplitude imagery is governed by multiplicative coherent speckle, whose negative-exponential single-look and Gamma L-look intensity statistics are classical [17], [18]. Constant-false-alarm-rate (CFAR) detection [19] and its truncated and cell-averaging variants are the canonical high-recall, low-precision ship locators built on these statistics, encoding the domain knowledge that such priors *localize* but do not *classify*. We position our multi-scale nonparametric evidence (MNE) prior squarely in this lineage rather than as a novel discriminator. Using a distribution-free rank statistic [20], we *quantify* its per-pixel ship/clutter behaviour and report the result against the *strongest* classical control, not a weak one: the single rank channel we actually deploy trails truncated and cell-averaging CFAR on inshore PSeg AUROC, and although a multi-scale fusion of the same statistic is in fact the best inshore discriminator we measured (a margin we disclose rather than hide), its edge falls below our pre-registered gate, so we ship the single zero-parameter channel for its cross-task sharing and its *recall* value, not for per-pixel classification. This honest negative is consistent with the in-principle non-separability of ships from man-made shore clutter under second-order statistics, and we guard every downstream use of the prior with a trivial-prior (random-seed) control, which confirms that a generic high-recall proposal does not substitute for the specific statistic.

D. Data leakage and segmentation-evaluation protocols

Train/test leakage is a recognized, pervasive threat to ML validity, and chip-tiling of a few large scenes is a natural vector in SAR: HRSID is cropped from large scenes at 25% overlap [2], so adjacent tiles can share imagery, yet to our knowledge no prior SAR segmentation work has audited the *official* split's contamination. We quantify it—92.7% of test chips share at least one pixel with a training chip (an upper bound on leakage), of which 10.8% overlap by $\geq 50\%$ area and exactly one is a near-duplicate—and release a leakage-free scene-level re-split (3659 train / 1945 test), reporting the official-vs-scene gap. We claim only the *quantification and repair*, not the discovery that chip tiling can leak, which follows from the documented chipping procedure. Relatedly, Boundary IoU [15] was introduced with the caveat that its dilation band approaches Mask IoU as objects shrink; we measure *how severe* this degeneration is in the SAR small-ship regime—the standard 2%-of-diagonal band collapses to Mask IoU for 91.5% of HRSID and 67% of PSeg instances—so a single-d Boundary IoU is not an independent quality axis here, and we report a multi-d curve. Both protocol findings stand independently of whether the evidence prior is strong.

E. SAR ship detection benchmarks

SAR ship *detection* is comparatively mature, with box mAP_{50} already near 0.99 on SSDD and 0.95 on HRSID. DETR-family detectors are the most relevant same-family baselines for SEAL's detection track, since SeedDINO injects evidence-derived seeds into a DINO [21] decoder's reference points, applying the established anchor-as-query lineage rather than proposing a new query formulation. Because the from-scratch-vs-COCO-pretrained protocol is itself unsettled in SAR, any comparison must declare each model's pretraining protocol explicitly. Accordingly, SEAL does *not* contest mAP_{50} leadership on this saturated benchmark. An oracle seed (ground-truth boxes) raises inshore query-level recall from 0.474 to 0.994, localizing the detection bottleneck at the query level; our actual contribution is the training-free, zero-parameter recall increment the shared statistical prior provides (two-seed-robust AR@100 +1.85 and small-object recall +2.1, with mAP and precision unharmed), and a

random-seed control confirms the increment is specific to the prior rather than to any high-recall proposal—the detection half of SEAL's two-track delivery.

III. Method

SEAL rests on a single methodological commitment: rather than proposing another architectural enhancement for the inshore regime, we first *diagnose* where the inshore AP_{75} gap originates—by a model-agnostic, AP-additive oracle decomposition—and only then apply the most economical surgery to the implicated half. The diagnose-then-fix posture has a natural-image lineage in error-decomposition diagnostics and oracle-driven refinement [11], [16]; our contribution is its application to inshore SAR small-ship instance segmentation, together with a specific decomposition that, to our knowledge, has no prior analogue in this domain. The diagnosis attributes the inshore AP_{75} deficit to two roughly equal halves: a *mask-quality* half (given a correct detection, the predicted mask falls short of the high-IoU criterion) and a *detection-coverage* half (the right instances are never proposed). A shared, training-free coherent-speckle evidence field supplies a high-recall prior consumed by both remedies, making the system a genuine dual delivery rather than two unrelated models. We present the components in the order they enter this argument: the oracle decomposition (Sec. III-A), the lightweight learned mask-refinement head and its equal-parameter control (Sec. III-B), the shared evidence prior (Sec. III-C), the evidence-seeded detector and inference-time mask-NMS (Sec. III-D), and the evaluation protocol (Sec. III-E).

A. AP-additive oracle decomposition of the inshore gap

We localize the inshore AP_{75} gap with a real, AP-additive oracle on the Mask2Former-R50 carrier [10], not a count heuristic: every oracle figure is a substitution-and-re-evaluation *ceiling*, not a score our system claims. Starting from the baseline inshore AP_{75}, we form two successive interventions, each re-running COCO evaluation after substituting ground truth along one axis while holding the other fixed.

Mask-quality oracle. Holding detection fixed, for every correctly detected instance ($\text{IoU} \geq 0.5$) we replace its predicted mask with the matched ground-truth mask and re-evaluate. Inshore AP_{75} rises from 0.341 to 0.637: perfecting the masks of already-detected instances recovers 0.296 AP_{75}.

Detection-coverage oracle. Adding back the missed ground-truth instances with perfect masks lifts AP_{75} further to 0.921, a 0.284 contribution from coverage. The two contributions are additive by construction, so the inshore deficit splits into $\sim$51% mask quality and $\sim$49% detection coverage; offshore the same procedure attributes $\sim$87% to mask quality, since offshore detection is near-complete. The headline diagnostic conclusion is therefore that mask quality is the *larger* inshore half—contradicting the intuition that inshore failures are dominated by missed detections—and is a concrete, addressable bottleneck that orientation-, query-, and prompt-based SAR designs leave untouched [6], [8], [9].

Detection-coverage is query-level, not feature-level. To verify that the coverage half is reachable by better proposals rather than blocked by feature inseparability, we inject ground-truth boxes as detection seeds into the reference points of a DINO carrier [21]. Inshore recall rises from 0.474 to 0.994 (a near-degenerate ceiling, reported only as such): given correct positions the classifier classifies correctly, so the inshore coverage bottleneck is query-level recall, not feature-level discriminability. This motivates the evidence-seeded detector of Sec. III-D.

Size-resolved mask head-room and its mechanism. Stratifying the mask-quality oracle by object size sharpens the prescription. The recoverable AP_{75} is $+0.253$ for small and $+0.394$ for medium instances (large instances are already saturated, $+0.000$); the medium head-room is thus the *larger*. The refinement of Sec. III-B, however, recovers only the small subset (AP_s $+1.2$, $+15$ instances crossing $\text{IoU} \geq 0.75$) and is inert on medium (AP_m -0.2, zero crossings). We read this precisely: at the coarse $1/4$-stride mask feature shared by all queries [10], a 19-pixel ship is described by $\sim$5 mask pixels, so the small-object head-room is *resolution-limited* and recoverable by adding mask capacity, whereas the (larger) medium-object head-room reflects shape and boundary detail that our remedy does not reach and that we flag as honest future work requiring a different lever. We do *not* claim that recoverable mask quality grows monotonically toward smaller objects.

B. SealM2FHead: lightweight learned mask refinement

The diagnosis prescribes added mask capacity at the final prediction. We realize it as *SealM2FHead*, a drop-in modification of the carrier that augments *only the final decoder layer's* mask, leaving the backbone, pixel decoder, the transformer decoder's nine auxiliary layers, the query set, and the bipartite matching unchanged. Restricting the change to the last layer is a deliberate, memory-bounded choice for single-GPU practice (one RTX 3090 Ti): an earlier attempt that upsampled the full pixel embedding to input resolution across all layers was both non-convergent and confounded with other changes, whereas refining only the final-layer mask is a clean, single-variable lever.

In standard Mask2Former, each query produces a low-resolution mask by a dot product between its mask embedding and the $1/4$-stride mask feature. For the N queries with embeddings $\mathbf{Q} \in \mathbb{R}^{N \times C}$ and the pixel-decoder feature $\mathbf{F}_{1/4} \in \mathbb{R}^{C \times \frac{H}{4} \times \frac{W}{4}}$, the per-query logits are

$$\mathbf{M}_{1/4} \;=\; \mathrm{einsum}\big(\text{“}nc,\, chw \to nhw\text{”};\; \mathbf{Q},\, \mathbf{F}_{1/4}\big)\,. \qquad (1)$$

The core of SealM2FHead is a lightweight residual refinement convolution $\mathcal{R}_\theta$ applied to the final-layer mask logits. We instantiate it in two variants that, together, disentangle *capacity* from *resolution*:

$$\textsf{caps}\ \text{(no resolution lift)}: \quad \mathbf{M}^{\mathrm{caps}} = \mathbf{M}_{1/4} \,+\, \mathcal{R}_\theta\big(\mathbf{M}_{1/4},\, \mathbf{F}_{1/4}\big)\,, \qquad (2)$$

$$\textsf{refine}\ (1/4 \to 1/2): \quad \mathbf{M}^{\mathrm{refine}} = \mathcal{U}\big(\mathbf{M}_{1/4}\big) \,+\, \mathcal{R}_\theta\big(\mathcal{U}\big(\mathbf{M}_{1/4}\big)\,,\, \mathbf{F}_{1/2}\big)\,, \qquad (3)$$

where $\mathcal{U}$ is a learned $\times 2$ upsampling and $\mathbf{F}_{1/2}$ the $1/2$-stride mask feature. The caps variant applies the *same* convolution at

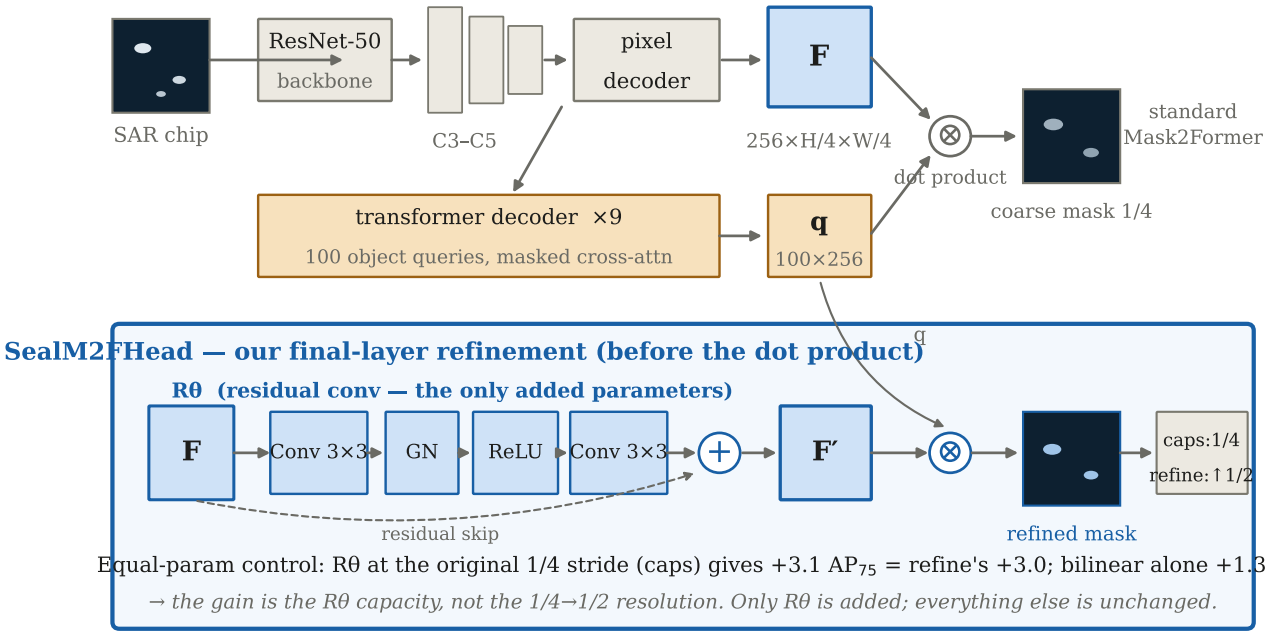


Fig. 2. SealM2FHead architecture. The backbone and pixel decoder produce the mask feature F at $1/4$ stride; a nine-layer transformer decoder produces the query embeddings q; the standard Mask2Former mask is the dot product $\langle q, F\rangle$. SealM2FHead inserts a small residual conv module R_θ (Conv 3×3–GroupNorm–ReLU–Conv 3×3) that refines F into F' *before* the final dot product; the *caps* variant keeps the $1/4$ stride while *refine* upsamples to $1/2$. R_θ is the only added parameter; an equal-parameter control shows the $+3.0$ AP_{75} gain comes from this added capacity, not from the resolution lift.

the original $1/4$ stride with no resolution increase and identical parameter count; refine additionally upsamples. The training loss, matching, and auxiliary-layer supervision are inherited unchanged; only the final-layer mask is modified.

The gain is added refinement capacity, not resolution. Because caps and refine differ *only* in whether the resolution is lifted, comparing them isolates the mechanism—a control that a resolution-only or learning-only ablation cannot provide, and which the literature on mask refinement typically skips. On PSeg-SSDD, caps reaches AP_{75} 0.643 on both seeds, identical to refine (0.643), while a bilinear-only control (the $1/4\rightarrow1/2$ resolution increase with $\mathcal{R}_\theta$ disabled) reaches only $+1.3$. The convolution applied at the original stride therefore captures the entire gain, and the upsampling contributes nothing on top of it. We conclude—and this is the central honest attribution of the paper—that the improvement comes from the added *refinement capacity* at the final layer, with resolution a minor, largely redundant contributor; we do not call this a "resolution axis" or claim that mask resolution is the binding constraint we fix. As a bonus, on PSeg-SSDD caps matches refine on AP_{75} with slightly better AP_m and no upsampling memory cost; we would therefore prefer it, though this rests on PSeg's small medium/large-instance counts and awaits the HRSID caps control (currently training) to confirm cross-dataset.

Headline result. Across three seeds, refine raises overall AP_{75} from a stock mean of 0.612 ($\{0.615, 0.623, 0.598\}$) to 0.642 ($\{0.643, 0.646, 0.638\}$), a $+3.0$ improvement that is positive on every seed ($+2.3$ to $+4.0$); caps is statistically indistinguishable (mean 0.643). The inshore AP_{75} improvement is larger at $+4.6$ (seed 0, $0.341 \rightarrow 0.387$) and lands exactly on the diagnosed mask-quality half, but an image-level paired bootstrap over the small 46-image inshore stratum (2000 resamples) gives a 95% CI of $[-0.5, +9.5]$ with $P(\Delta > 0) = 0.96$—suggestive, not significant—so we headline the seed-robust overall $+3.0$ and report the inshore figure with its interval. We do *not* claim the overall-AP ($+0.4$) or AP_s ($+0.5$) deltas, which lie within seed noise, and we say so plainly. We position the coarse-to-fine refinement *paradigm* as mature, not novel—RefineMask, PointRend, Mask Transfiner, and DynaMask establish it [11]–[13], [22], and our $+3.0$ sits within their reported range—so our contribution on this axis is the diagnosis that final-layer mask capacity is a binding inshore SAR constraint, its minimal single-GPU instantiation, and the equal-parameter control that correctly attributes the gain.

C. Multi-scale nonparametric evidence (MNE) field

SAR amplitude imagery is governed by multiplicative coherent speckle: under the Goodman random-walk model, single-look intensity is negative-exponentially distributed and an L-look average follows a Gamma law with mean one and variance $1/L$ [17], [18]. A fixed global threshold degrades sharply in cluttered inshore scenes, which is why the classical detection lineage adopts locally adaptive, constant-false-alarm-rate thresholding [19]. In that spirit we compute a training-free

multi-scale nonparametric evidence (MNE) field that is locally adaptive and distribution-free. For a pixel p and a set of scales $\mathcal{S}$, we form a right-truncated rank statistic of the test pixel against a local reference annulus and aggregate across scales,

$$E(p) \;=\; \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}} \Phi\Big(\,\mathrm{rank}_s\big(I(p)\,;\,\mathcal{R}_s(p)\big)\Big), \qquad (4)$$

where $I(p)$ is the (log-)amplitude, $\mathcal{R}_s(p)$ the local clutter reference at scale s, $\mathrm{rank}_s(\cdot)$ the right-truncated rank of the test pixel within that reference (the truncation suppresses target self-contamination of the reference set), and Φ a monotone normalizer. The rank statistic is the two-sample Mann–Whitney form [20], making no parametric assumption on the local clutter beyond the Gamma family above. The field is equivariant to geometric augmentation, so we precompute it once per image and transform its derived seeds together with the image. From E we derive a high-recall set of seed boxes (consumed by Sec. III-D) and an optional per-pixel evidence channel.

An honest characterization: a recall prior, not a classifier. We report what the prior is and is not, against the *strongest* classical control rather than the weakest. As a per-pixel ship/clutter discriminator measured by AUROC, the single rank channel we deploy (MNE-E) is *not* the best: on inshore PSeg-SSDD it trails the truncated- and cell-averaging CFAR statistics (0.928 vs 0.950 and 0.952), and on HRSID a raw box statistic leads (0.992 vs 0.986). A multi-scale fusion of the evidence (MNE-FULL) *is* the best inshore discriminator (0.960; $+1.0$ over the strongest CFAR control and $+7.6$ over a box z-score), but we deliberately deploy only the single rank channel—for zero added parameters, cross-task sharing between detection and segmentation, and because the fusion's discrimination margin falls below our pre-registered $+0.03$ deployment gate. We disclose this fusion edge explicitly rather than comparing only against the weakest control. The value of the deployed prior is therefore *recall*, not per-pixel classification, consistent with the in-principle non-separability of ship and man-made shore-clutter statistics in single-look amplitude—the same fact that motivates the two-stage "locate then discriminate" pattern of the CFAR lineage [19].

D. SeedDINO evidence seeding and inference-time mask-NMS

SeedDINO. The coverage half of the diagnosis is query-level recall (Sec. III-A), with an oracle ceiling of 0.994 inshore recall under perfect seeds. SeedDINO tests how much a real, training-free prior recovers by injecting MNE-derived seed boxes into the DINO decoder's reference points [21]; because seeds enter as reference points rather than encoder features, the spatial prior survives the per-layer positional-encoding refresh and reaches the final prediction, as the oracle confirms. On PSeg-SSDD across two seeds, predicted-seed injection lifts AR@100 by $+1.85$ (per-seed $+1.8/+1.9$) and small-object recall by $+2.1$, at flat mAP (mean $+0.15$; seed-0 $+0.4$, seed-1 -0.1). We keep the observed precision rise out of our causal claims: it is single-seed and confounded with a confidence-threshold shift. Critically, because a recall gain could in principle come from *any* high-recall proposal rather than from the speckle statistic specifically, we run a trivial-prior control: random seeds give bbox mAP 0.631, *below* both no-seed (0.638) and MNE-seed (0.642), so random seeding does not help (it slightly hurts) and the MNE statistic is specifically useful, not interchangeable with arbitrary proposals. We position SeedDINO as the training-free, zero-parameter detection half of the dual delivery, not a standalone detector, and do not contest mAP_{50} on this saturated benchmark.

Inference-time mask-NMS. Independently of training, a standard overlap-based mask non-maximum suppression applied at inference reduces the dominant inshore failure mode—instance fragmentation (split)—at unchanged AP. It lowers the split rate from 0.344 to 0.152 on PSeg-SSDD and from 0.303 to 0.108 on HRSID, while being nearly inert on merges. We present mask-NMS as a cheap, zero-training post-processing tool, not a methodological contribution.

E. Evaluation protocol

Two protocol elements are part of the method because they determine what the reported numbers mean; both stand independently of the strength of the evidence prior.

Leakage-free scene-level split. HRSID [2] is tiled with overlap from a small number of large source scenes, so chips from one scene can populate both partitions of the de-facto official split. Auditing each test chip's maximum same-scene pixel overlap with any training chip, we find 92.7% of test chips share at least one pixel with a training chip—an *upper bound* on leakage—of which 10.8% overlap by $\geq 50\%$ of their area and exactly one is a full duplicate. We therefore report the full severity distribution rather than a single alarming percentage, and construct a leakage-free *scene-level* re-split that partitions by source scene (3659 train / 1945 test chips, no cross-split scene membership). We claim the quantification and the repair, not the unsurprising fact that chip tiling can leak. HRSID results reported on the official split (A0 mask AP 0.348) are thus optimistic and are labeled as such.

Multi-d Boundary IoU. Boundary IoU [15] was introduced with the caveat that its dilation band approaches Mask IoU as objects shrink; for the extremely small ships dominating SAR data this caveat becomes the rule. Measuring each image against its own diagonal, the standard 2%-of-diagonal band degenerates to plain Mask IoU for 91.5% of HRSID and 67% of PSeg instances, so a single-d Boundary IoU is not an independent quality axis in this regime. We therefore report a multi-d curve and treat the fine $d = 2\,\mathrm{px}$ band as the only boundary-discriminative operating point.

Cross-dataset stress test. We use the dense HRSID scenes as a stress test of the refinement head rather than a second headline, and report all HRSID deltas as single-seed without a significance claim (we have no HRSID seed-variance estimate and do not import the PSeg one). On the official split the head is neutral on AP_{75} ($0.378 \rightarrow 0.383$, $+0.5$, within single-seed noise) and does no harm. A per-instance mechanism analysis, however, shows that matched mask-IoU *drops* by 0.027: the flat-to-slightly-positive aggregate is attributable to recall re-ranking, not to improved mask quality, so the refinement capacity does *not* transfer to the dense HRSID regime,

TABLE II
CARRIER BASELINES (SINGLE RTX 3090 TI). A0 (MASK2FORMER-R50) IS THE SHARED CARRIER FOR THE ORACLE DIAGNOSIS AND THE REFINEMENT HEAD; DINO-R50 IS THE DETECTION CARRIER. HRSID VALUES ARE ON THE OFFICIAL (LEAKY) SPLIT; SEE SEC. IV-G.

Carrier	Dataset / metric	Value
A0 (Mask2Former-R50)	PSeg-SSDD segm AP	0.518
A0 (Mask2Former-R50)	PSeg-SSDD segm AP_{75} (mean, 3 seeds)	0.612
A0 (Mask2Former-R50)	HRSID segm AP / AP_{50}	0.348 / 0.625
DINO-R50	PSeg-SSDD box mAP	0.638

TABLE III
REFINEMENT-HEAD MAIN RESULT ON PSEG-SSDD, THREE SEEDS PER CONFIGURATION. THE OVERALL AP_{75} GAIN (MEAN $+3.0$) IS SEED-ROBUST: EVERY PER-SEED DELTA IS POSITIVE ($+2.3$ TO $+4.0$). OVERALL AP AND AP_s ARE WITHIN SEED NOISE AND ARE NOT CLAIMED.

Metric	A0 (mean / seeds)	A0R-refine (mean / seeds)	mean Δ
AP_{75}	0.612 / {.615,.623,.598}	0.642 / {.643,.646,.638}	**+3.0**
AP	0.518	0.522	+0.4 (noise)
AP_s	0.499	0.504	+0.5 (noise)

AP and AP_s are 3-seed means; the AP_s per-seed spread (A0 $\{.494, .511, .489\}$, A0R $\{.506, .509, .498\}$) exceeds the $+0.5$ mean delta, so it is within seed noise and not claimed.

and we do not claim that mask cleanliness generalizes. We report the fragmentation change (split $0.303 \rightarrow 0.288$, merge $0.153 \rightarrow 0.141$) explicitly as a count change with the caveat that per-instance IoU did not improve. Finally, on HRSID's 76 *real* large instances the large-object AP is essentially flat ($0.126 \rightarrow 0.124$, -0.2); this is *consistent with* the within-PSeg large-object delta (-13.8 at seed 0, $0.7134 \rightarrow 0.5757$) being a small-sample artifact of that split's two large instances—which sign-flips to $+2.7$ at seed 1—rather than a regression we have dispelled.

IV. EXPERIMENTS

We organize the evaluation around the diagnostic claim itself: that the inshore AP_{75} gap of a strong query-based carrier splits into a detection-coverage half and a mask-quality half, and that a *minimal, honestly-attributed* surgery recovers part of the implicated share under a fixed training budget. We report, in order, the setup (Sec. IV-A), the refinement-head main result with its *equal-parameter* ablation that is the centerpiece of the rigor (Sec. IV-B), the inshore slice with its bootstrap CI (Sec. IV-C), the honest HRSID cross-dataset check (Sec. IV-D), the real AP-additive oracle decomposition (Sec. IV-E), the detection-seeding result with its trivial-prior control (Sec. IV-F), the evaluation-protocol contributions (Sec. IV-G), and the honest negative results that bound every claim, including the per-pixel-discrimination panel (Sec. IV-H).

A. Datasets, Carriers, Metrics, and Protocol

Datasets. We use PSeg-SSDD [1], the polygon-segmentation annotation of the official SSDD, under its official split with the test set stratified into inshore/offshore by the dataset's own definition; and HRSID [2] (MS-COCO-format masks). The inshore/offshore stratification originates with the datasets, not with us; we adopt it so that our per-slice numbers align with the protocol used by competing methods [8], [9].

Carriers. The segmentation carrier (denoted A0) is Mask2Former-R50 [10] (mmdet 3.3.0) trained for 36 epochs on PSeg-SSDD and for ∼100k iterations on HRSID. The detection carrier is DINO-R50 [21], into whose decoder reference points we optionally inject evidence seed boxes (SeedDINO). Table II lists the carrier anchors that ground all subsequent deltas. All runs use a single RTX 3090 Ti.

Metrics. We report COCO mask AP, AP_{50}, AP_{75}, and the size-stratified AP_s/AP_m/AP_l, under inshore/offshore stratification, together with split and merge rates as measures of mask fragmentation. We additionally report a multi-distance Boundary IoU [15]; we quantify its degeneration in the SAR small-ship regime in Sec. IV-G.

B. Main Result: Lightweight Learned Mask Refinement, Capacity not Resolution

The head. SealM2FHead applies a small convolutional module R_θ that refines the final decoder layer's mask embedding, leaving the nine auxiliary layers and the bipartite matching unchanged. In its default form (A0R-refine) the refined embedding is also upsampled from the standard $1/4$ stride to $1/2$. Restricting the change to the final layer keeps memory tractable on a single GPU and makes the experiment a clean single-variable test.

Seed-robust headline. To prevent single-seed over-claiming we run *three* seeds for both A0 and A0R-refine on PSeg-SSDD (Table III). The overall AP_{75} gain is robust: stock mean 0.612 ($\{0.615, 0.623, 0.598\}$) versus refine mean 0.642 ($\{0.643, 0.646, 0.638\}$), a mean delta of **+3.0** with every per-seed delta positive ($+2.3$ to $+4.0$). By contrast the overall AP ($+0.4$) and small-object AP_s ($+0.5$) deltas lie within seed noise, and we explicitly do *not* claim them.

Equal-parameter control: the gain is capacity, not resolution. The decisive ablation answers a reviewer's first question—does the gain come from the $1/4 \rightarrow 1/2$ resolution lift, or from the added refinement module? We isolate the two factors with two controls (Table IV). A *bilinear-only* control (pure resolution, no convolution) yields only $+1.3$ AP_{75}. An *equal-parameter* control, **A0R-caps**, applies the *same* convolutional module R_θ at the *original* $1/4$ stride with *no* resolution increase, and reaches AP_{75} 0.643—matching A0R-refine's 0.642 and reproducing on both available seeds. The conclusion is unambiguous: the gain comes from the added refinement *capacity* at the final layer, not from the resolution increase, which is a minor and largely redundant contributor. We therefore frame the method as *lightweight learned mask refinement*, and we do not call it a resolution axis. As a practical bonus, A0R-caps slightly improves AP_m/AP_l and incurs no upsampling memory cost, making it a candidate configuration to ship.

This refinement *paradigm* is itself mature [11]–[13]; our $+3.0$ AP_{75} sits inside the range those methods report (Fig. 3).

TABLE IV
EQUAL-PARAMETER ABLATION ON PSEG-SSDD. PURE RESOLUTION (BILINEAR, NO CONV) GIVES ONLY +1.3; THE EQUAL-PARAMETER CONTROL A0R-CAPS—THE SAME CONV MODULE AT THE *original* 1/4 STRIDE, NO RESOLUTION LIFT—MATCHES THE FULL A0R-REFINE. THE GAIN IS THEREFORE ADDED REFINEMENT *capacity*, NOT RESOLUTION.

Configuration	resolution	conv R_θ	AP_{75}
A0 (stock)	1/4	–	0.615
A0R-bilinear	1/2	–	0.628 (+1.3)
A0R-caps (equal-param)	1/4	✓	0.643 (+2.8)
A0R-refine (default)	1/2	✓	0.643 (+2.8)

All rows are seed-0 for a single consistent baseline; caps reproduces 0.643 on seed-1. The conv at the original 1/4 stride (caps) recovers the full gain—equal to refine, far above bilinear—so the gain is added capacity, not resolution. The seed-robust headline (+3.0 AP_{75} for refine over three seeds) is in Table III.

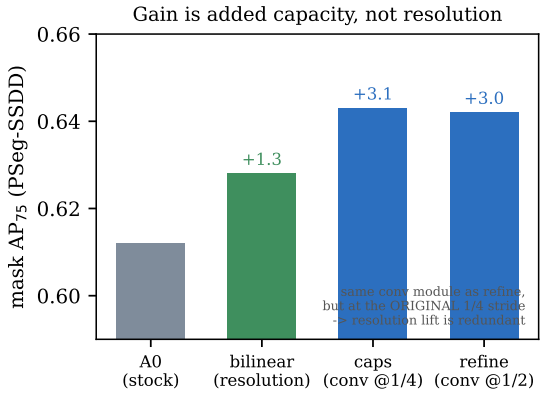


Fig. 3. The refinement gain is added *capacity*, not resolution. An equal-parameter control (caps: the same conv module R_θ at the original 1/4 stride, with no resolution lift) matches the full-resolution refine model (+3.1/+3.0 AP_{75}), whereas a bilinear-only control (pure resolution) yields only +1.3. PSeg-SSDD; stock is the 3-seed mean.

Our contribution is not the paradigm but the diagnosis that motivates a final-layer-only instantiation, and the equal-parameter control that honestly attributes the gain. We further note the gain is in high-IoU area coverage, not at the finest boundary: the $d = 2$ px Boundary IoU is essentially flat, so we claim "learned refinement", not "sharper boundaries".

C. Inshore Slice

The inshore AP_{75} delta is larger than the overall figure (+4.6 at seed 0, 0.341 → 0.387)—the slice the diagnosis (Sec. IV-E) implicates—but it is statistically weaker on the small 46-image inshore stratum. A paired image-level bootstrap (2000 resamples over the 46 inshore images) places it at a 95% CI of $[-0.5, +9.5]$, which includes zero, with $P(\Delta > 0) = 0.96$. We therefore report the inshore +4.6 as *suggestive and consistent with the diagnosis, but not significant*, and headline the seed-robust overall +3.0 instead. The refinement also targets the resolution-limited small-object subset specifically: it improves AP_s (with +15 instances crossing the IoU ≥ 0.75 threshold) but not AP_m (−0.2, zero net crossings), exactly the pattern the size-resolved oracle predicts in Sec. IV-E.

TABLE V
CROSS-DATASET REFINEMENT-HEAD RESULT ON HRSID (OFFICIAL SPLIT, SINGLE SEED; ITER 100848, LR≈ 0). THE AGGREGATE IS NEUTRAL AND REPORTED *without* A SIGNIFICANCE CLAIM (NO HRSID SEED-VARIANCE ESTIMATE). A PER-INSTANCE ANALYSIS FINDS MATCHED MASK IOU DROPS −0.027: THE FLAT AGGREGATE IS RECALL RE-RANKING, NOT MASK-QUALITY IMPROVEMENT—THE MECHANISM DOES NOT TRANSFER. AP_l ON 76 REAL LARGE INSTANCES (−0.2) IS CONSISTENT WITH THE PSEG −13.8 BEING A TWO-INSTANCE ARTIFACT.

Metric	A0	A0R-refine	Δ
AP	0.348	0.351	+0.3
AP_{50}	0.625	0.633	+0.8
AP_{75}	0.378	0.383	+0.5 (neutral)
AP_s	0.339	0.344	+0.5
AP_m	0.466	0.448	−1.8 (caveat)
AP_l (76 inst.)	0.126	0.124	−0.2 (consistent with)
matched mask IoU	per-instance		−0.027
split rate	0.303	0.288	−0.015
merge rate	0.153	0.141	−0.012

D. HRSID Cross-Dataset: Neutral, and the Mechanism Does Not Transfer

We apply the head to HRSID (official split, converged model; iter 100848, learning rate ≈ 0) as an honest stress test (Table V). The aggregate is *neutral*: AP_{75} +0.5 (0.378 → 0.383), reported as a single-seed delta *without* a significance claim, since we have no HRSID seed-variance estimate and do not import the PSeg σ. Crucially, a per-instance mechanism analysis shows that the matched mask IoU *drops* by −0.027: the flat-to-slightly-positive aggregate is attributable to recall re-ranking, *not* to improved mask quality. We therefore state plainly that the refinement *does not transfer* to the dense HRSID regime, and we do *not* claim that mask cleanliness generalizes. The fragmentation counts move (split 0.303 → 0.288, merge 0.153 → 0.141), but we report this as a count change with the explicit caveat that per-instance IoU did not improve.

The HRSID large-object axis serves a separate purpose: on its **76 real large instances**, AP_l moves only 0.126 → 0.124 (−0.2). This is *consistent with* the apparent PSeg-SSDD large-object delta (−13.8, 0.7134 → 0.5757 at seed 0) being a two-instance small-sample artifact—PSeg's test set contains only two large instances—rather than a real regression; the within-PSeg sign flip across seeds (seed 0 −13.8, seed 1 +2.7) is the primary noise evidence. We say "consistent with", not "dispels". The leakage-free scene-split refinement run is in progress [TODO: HRSID scene-split refine result; caps-HRSID equal-parameter control].

E. Diagnosis: Real AP-Additive Oracle Decomposition

We localize the inshore gap with a *real, AP-additive* oracle: holding detection fixed, we swap the masks of correctly-detected instances for their ground-truth masks and re-run COCOeval, then add the masks of all undetected ground-truth instances (Table VI). On inshore PSeg-SSDD, AP_{75} rises 0.341 → 0.637 (perfect masks for detected instances) → 0.921 (perfect masks plus full detection coverage). This decomposes the inshore gap into roughly **51% mask quality** and **49% detection coverage**; on offshore the mask share is

TABLE VI
REAL AP-ADDITIVE ORACLE DECOMPOSITION ON PSEG-SSDD INSHORE. DETECTION IS HELD FIXED; DETECTED-INSTANCE MASKS ARE REPLACED BY GROUND TRUTH (MASK ORACLE), THEN FULL GROUND-TRUTH COVERAGE IS ADDED (FULL ORACLE). THE INSHORE AP_{75} GAP IS $\sim$51% MASK QUALITY / $\sim$49% DETECTION COVERAGE; MASK IS THE LARGER INSHORE HALF (OFFSHORE: 87% MASK).

Stage	inshore AP_{75}	recovered share
A0 baseline	0.341	–
+ mask oracle (detected)	0.637	$\sim$51% (mask)
+ full GT coverage	0.921	$\sim$49% (detection)

Size-resolved mask-recoverable AP_{75}: small $+0.253$, medium $+0.394$, large $+0.000$. The refinement recovers only the small (resolution-limited) band.

TABLE VII
DETECTION SEEDING ON PSEG-SSDD. TOP: MNE SEEDING GIVES A SEED-ROBUST RECALL GAIN (AR@100 $+1.85$) AT FLAT MAP (TWO SEEDS). BOTTOM: THE TRIVIAL-PRIOR CONTROL—RANDOM SEEDS GIVE MAP $0.631 <$ NO-SEED $0.638 <$ MNE-SEED 0.642—SHOWS RANDOM SEEDING DOES NOT HELP; THE MNE STATISTIC IS SPECIFICALLY USEFUL. ORACLE (GT-BOX) RECALL $0.474 \rightarrow 0.994$ IS A CEILING.

Configuration	AR@100	bbox mAP
DINO no-seed (s0/s1)	0.742 / 0.736	0.638
+ MNE seed (s0/s1)	0.760 / 0.755	0.642
+ random seed (control)	–	0.631
+ GT-box oracle	recall 0.994	ceiling

Oracle GT-box seeding lifts inshore recall $0.474 \rightarrow 0.994$, locating the bottleneck at the query level.

87%. The mask half is thus the *larger* inshore component, which strengthens—rather than weakens—the motivation for a mask-side remedy.

A size-resolved version of the same oracle reports the mask-recoverable AP_{75} head-room per size band: small $+0.253$, medium $+0.394$ (the *larger* pool), large $+0.000$. Critically, the refinement recovers *only* the small band (AP_s $+1.2$, $+15$ instances crossing IoU ≥ 0.75) and *not* the medium band (AP_m -0.2, zero net crossings). The interpretation is honest and specific: the refinement addresses the *resolution-limited* small-object subset, whereas the larger medium-object mask head-room is *not* resolution-limited and remains open future work. We make no claim that recoverable head-room grows monotonically for smaller objects.

F. Detection Seeding: a Training-Free Recall Booster with a Trivial-Prior Control

A complementary, training-free component injects the high-recall MNE seed boxes into the DINO decoder's reference points (SeedDINO), positioned as a *zero-parameter* recall booster, not an independent detector. An oracle that injects ground-truth boxes lifts inshore recall from 0.474 to 0.994, confirming that the inshore bottleneck sits at the query level (proposal coverage), not at feature-level discrimination—were the collapse feature-driven, AP could not approach 0.994 even given correct locations. With real predicted seeds, across two seeds (Table VII), seeding improves AR@100 by **+1.85** (per-seed $+1.8$/$+1.9$) and small-object AR by $+2.1$, while bbox mAP is flat (mean $+0.15$; seed 0 $+0.4$, seed 1 -0.1): added recall is offset at the AP operating point. A threshold-resolved analysis reveals the *mechanism*: the averaged-AR gain is driven by improved high-IoU localization (recall@0.9 is $+6.2$ pp) rather than loose-threshold coverage (recall@0.5 is in fact -1.3 pp overall and -3.5 pp inshore). In other words, the low-precision evidence seeds (17.6% seed precision) *tighten the localization* of objects the detector already finds rather than expanding coverage; the coverage bottleneck the oracle exposes—which perfect seeds close ($0.474 \rightarrow 0.994$)—thus remains open to imperfect ones, a concrete direction for higher-precision seeding. This honest mechanism is consistent with the random-seed control above.

Trivial-prior control: the statistic is specifically useful. A reviewer-required random-seed control isolates whether the *MNE statistic*—rather than any high-recall proposal source—drives the gain. Random seeds give bbox mAP 0.631, *below* the no-seed baseline 0.638, while MNE seeds give 0.642. Random seeding therefore does *not* help (it slightly hurts); the gain is specific to the MNE statistic, not to trivial high-recall injection. We deliberately keep the observed precision rise out of the causal claim: it is single-seed and confounded by a confidence-threshold shift. A random-seed *AR*-specific evaluation is pending [TODO: random-seed AR control; third detection seed in progress].

Mask-NMS (training-free, post-hoc). A training-free mask-NMS post-process substantially reduces the split rate—PSeg-SSDD $0.344 \rightarrow 0.152$, HRSID $0.303 \rightarrow 0.108$—while leaving AP unchanged. We present it as a cheap diagnostic/post-processing tool that quantifies fragmentation as a dominant inshore structural failure mode, not as a methodological contribution.

G. Evaluation Protocol: Leakage and Boundary-IoU Degeneration

Official-split leakage. HRSID chips are cropped from a few large source scenes with substantial inter-chip overlap. Auditing the de-facto (official) split, we find that **92.7%** of test chips share at least one pixel with a same-scene training chip—an *upper bound* on leakage—while only **10.8%** overlap by $\geq 50\%$ of their area (the severe regime) and exactly one chip is a near-complete duplicate. We make the narrow claim that we *quantify* this severity distribution, not that we discovered the overlap. To enable leakage-free evaluation we release a reproducible **scene-level re-split** that partitions source scenes (not chips) into disjoint groups, yielding **3,659 train / 1,945 test** chips with no cross-split scene membership. We emphasize that the HRSID anchors above are on the *official* (leaky) split and are therefore optimistic.

Boundary-IoU degeneration. The standard Boundary IoU [15] dilates by $d = 2\%$ of the image diagonal; on the extremely small ships that dominate SAR data this band engulfs whole instances, so Boundary IoU degenerates to Mask IoU. Measuring this directly on A0, the standard d makes Boundary IoU equal to Mask IoU for **91.5%** of HRSID and **67%** of PSeg-SSDD instances. We therefore report a multi-distance curve and treat the $d = 2$ px operating point as

TABLE VIII
PER-PIXEL SHIP/CLUTTER AUROC (G1), REPORTED AGAINST THE *strongest* CLASSICAL CONTROL. THE SHIPPED SINGLE RANK CHANNEL MNE_E IS NOT THE BEST DISCRIMINATOR; THE UNSHIPPED FUSION MNE_FULL LEADS INSHORE PSEG (+1.0 OVER TS_CFAR). WE DEPLOY MNE_E FOR ZERO PARAMETERS AND CROSS-TASK SHARING, AND BECAUSE ITS MARGIN IS BELOW THE PRE-REGISTERED $+0.03$ GATE—ITS VALUE IS RECALL, NOT CLASSIFICATION.

Channel	PSeg inshore	HRSID (all)
mne_full (fusion, unshipped)	**0.960**	0.990
ca_cfar / ts_cfar	0.952 / 0.950	–
box_raw	–	**0.992**
mne_E (shipped, single channel)	0.928	0.986
box z-score (weakest)	0.884	–

the only boundary-discriminative one; the phenomenon itself is anticipated by [15], but the measured percentages are ours.

H. Honest Negative Results

Because the shared statistical prior is a supporting actor rather than the core contribution, we can afford a rigorous characterization of where it helps and where it does not. We regard these as a credibility asset, not method failures.

The prior is not the best per-pixel discriminator (G1). We treat the MNE field, built from coherent-speckle statistics [17], as a per-pixel ship/clutter discriminator and compare it against the *strongest* classical control (Table VIII), not a straw-man. The shipped single rank channel (mne_E) is *not* the best discriminator: on inshore PSeg-SSDD it trails ts_cfar (AUROC 0.928 vs. 0.950) and ca_cfar (0.952); on HRSID box_raw leads (0.992 vs. mne_E 0.986). A multi-scale fusion (mne_full) *is* the best inshore PSeg discriminator (0.960; $+1.0$ over ts_cfar, $+7.6$ over a box z-score), and we disclose this edge openly. We nonetheless deploy the single rank channel for its zero parameters, cross-task sharing, and because the discrimination margin is below our pre-registered $+0.03$ gate—its value is recall, not per-pixel classification. This is consistent with the in-principle non-separability of ships from man-made shore clutter by amplitude alone.

Within-noise non-claims. For completeness we restate the deltas we do *not* claim: overall AP ($+0.4$) and small-object AP_s ($+0.5$) on the refinement head are within seed noise; detection-seeding mAP is flat; HRSID AP_{75} ($+0.5$) is neutral and reported without a significance claim; the inshore $+4.6$ is suggestive with a CI including zero. Reported plainly, these bound the headline—a seed-robust $+3.0$ overall AP_{75} from added refinement *capacity*, with an equal-parameter control proving the mechanism—to what the evidence supports. We make no aggregate mask-AP state-of-the-art claim; the negative results sharpen, rather than weaken, the diagnosis.

V. CONCLUSION

We presented a diagnose-before-fix study of inshore small-ship instance segmentation in SAR imagery. Rather than adding capacity blindly, we first asked *where* the inshore high-IoU gap of a strong query-based carrier (Mask2Former-R50) originates, and answered it with a real, AP-additive oracle decomposition: holding detection fixed and substituting ground-truth masks for the detected instances, then adding back the missed instances, partitions the inshore AP_{75} deficit into roughly equal halves—$\sim$51% mask quality and $\sim$49% detection coverage (inshore $0.341 \to 0.637 \to 0.921$; offshore is $\sim$87% mask). Mask quality is thus the *larger* inshore half, which both sharpens the diagnosis and motivates the remedy.

Guided by this decomposition, our segmentation main line is a lightweight *learned* final-layer mask refinement: a small residual convolution that refines the final decoder layer's mask embedding, optionally upsampled. On PSeg-SSDD it yields a seed-robust $+3.0$ AP_{75} over three seeds (all per-seed deltas positive, $+2.3$ to $+4.0$). We attribute this gain honestly, and we regard the attribution itself—not a headline number—as a central rigor asset of the paper. An equal-parameter control that applies the *same* convolution at the original $1/4$ stride, with no resolution increase, matches the refined head's AP_{75} (0.643 on both seeds, versus 0.642 for the upsampled head), whereas a bilinear-only control that adds resolution without the convolution recovers only $+1.3$. The gain therefore comes from the added refinement *capacity* at the final layer, not from a $1/4 \to 1/2$ resolution lift; resolution is a minor, largely redundant contributor. An instance-level analysis confirms the mechanism lands where the diagnosis predicted: the refinement pushes resolution-limited small-object masks across the high-IoU threshold (AP_s $+1.2$, $+15$ instances crossing $\text{IoU} \geq 0.75$), while the larger medium-object mask headroom—which the oracle shows is *not* resolution-limited—is left for future work. An image-level bootstrap places the inshore $+4.6$ AP_{75} (seed 0) at a 95% CI of $[-0.5, +9.5]$, suggestive but not significant on the 46-image inshore stratum, so we headline the overall $+3.0$ and report the inshore effect with its interval.

We treated HRSID as an honest cross-dataset stress test rather than a second headline. There the refinement is neutral (AP_{75} $0.378 \to 0.383$, single-seed, within noise) and a per-instance analysis shows matched mask-IoU in fact *drops* (-0.027); the flat-to-slightly-positive aggregate is attributable to recall re-ranking, not to improved mask quality, so the resolution-limited mechanism does *not* transfer to dense HRSID scenes. On HRSID's 76 real large instances the large-object AP is flat (-0.2), which is *consistent with* the PSeg large-object delta (seed 0 -13.8, seed 1 $+2.7$) being a two-instance small-sample artifact rather than a regression.

Alongside the segmentation line, we gave a deliberately two-sided account of a training-free coherent-speckle statistical prior. As a per-pixel ship/clutter discriminator the shipped single rank channel is honestly *not* the best classifier—inshore it trails the strongest classical control (ts-CFAR), and a multi-scale fusion of the same statistic is actually the best inshore discriminator—yet we deploy the single channel for its zero parameters, cross-task sharing, and because its margin falls below our pre-registered gate; its value is recall, not per-pixel classification. Injected into a DETR detector's reference points, the prior gives a seed-robust $+1.85$ AR@100 ($+2.1$ small-object recall) at flat mAP, and a random-seed trivial-prior control confirms the *specific* statistic helps where arbitrary high-recall proposals do not (random seeding slightly hurts). Finally, our evaluation-protocol contributions stand

independently of the prior's strength: we quantify the substantial chip-level leakage in the de-facto HRSID split (92.7% any-overlap, 10.8% by $\geq 50\%$ area, one full duplicate) and release a leakage-free scene-level re-split, and we measure how the standard Boundary-IoU dilation degenerates to Mask-IoU for the extremely small ships that dominate SAR data (91.5% / 67% of HRSID/PSeg instances), motivating a multi-d report.

We deliberately make no aggregate mask-AP state-of-the-art claim; our contribution is a diagnostic framework with a capacity-attributed minimal remedy, a rigorously bounded study of a statistical prior, and a reproducible evaluation protocol. Three directions follow naturally. First, the medium-object mask headroom is the larger remaining inshore deficit and is *not* resolution-limited, calling for a shape- or boundary-aware mechanism rather than further resolution. Second, validating the refinement on the leakage-free HRSID scene split—and addressing the dense-scene mask bleed that our per-instance analysis exposed—would test the remedy under a clean protocol. Third, an adaptive, more strongly parameterized refinement is the path toward cross-dataset and private-data deployment, where the resolution-limited regime the current head targets is only one of several mask-quality bottlenecks.

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